

Coding Workshop

September 26th, 2023



Memorial Sloan Kettering
Cancer Center

Agenda

- **Hackathon Announcements (~10min)**
 - Caroline Kostrzewa
- **Review of CRM Designs (~15 min)**
 - Joanne Chou
- **Intro to Coding CRM Simulations & Coding Exercise (~30 min)**
 - Joanne Chou & Karissa Whiting

Hackathon Announcements

October 11 - 12

- [Hackathon - Fall 2023 Task List](#) – still being updated (we will update participants when it's completed)
 - Send us your ideas!
- [Resource Guide - Hackathon Cheat Sheet](#)
 - Refresh your memory on GitHub and PR best practices *before* coming to the Hackathon

Contact us (Caroline & Hannah F) if you have questions/comments/suggestions!

Simulations for Phase I CRM Clinical Trial Design

Joanne Chou & Karissa Whiting

Coding Workshop
September 26th, 2023

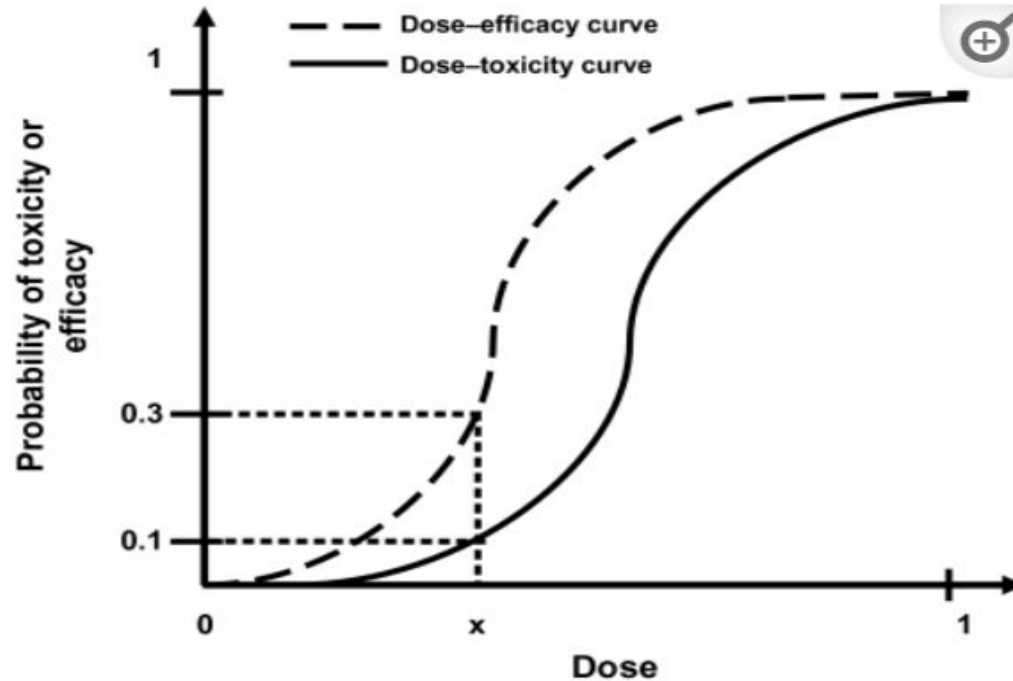


Memorial Sloan Kettering
Cancer Center

Quick Review: Phase I Design

- **Objective is to evaluate safety** and **feasibility** of a treatment regimen
- The objectives of the phase I design:
 - Determination of qualitative and quantitative **toxicity** and of the toxicity profile
 - Characterization of **Dose-Limiting Toxicity (DLT)**
 - Establishment of a **Maximum Tolerated Dose (MTD)**

Dose Toxicity & Dose efficacy



Rules based vs Model based

- **Pros**

- Transparent
- Easy to implement

- **Cons**

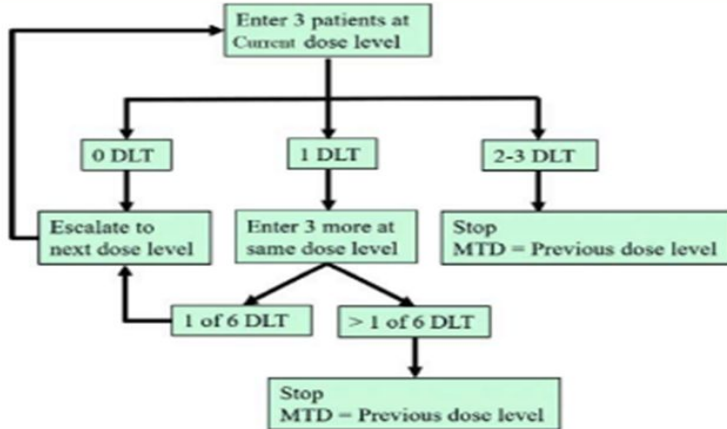
- Poor performance (many pts treated at subtherapeutic dose)
- Slower dose escalation
- Dose escalation solely dependent on current dose

- **Pros**

- Solid statistical foundation
- Flexible and efficient
- Better performance than 3+3
- Continuously updates the prob{toxicity} at various dose levels

- **Cons**

- Need to have a prior guess
- Computation after each cohort of patients
- Challenging to communicate with non-statisticians



Black Box

Model-Based Dose-Escalation Designs CRM

Clinical and Model Tuning Parameters must be specified at the time the protocol is written and before any patients are enrolled

Clinical parameters

- Prespecified **window of DLT** observation
- Types of adverse events **define DLT**
- **Cohort size**
- Safety stopping rules
- Predetermined **toxicity threshold** (eg 33%)

Model Tuning parameters

- Presumed **dose-toxicity relationship** (called the “initial curve”)
- Prior information from historical data

Operating Characteristics of CRM Models

Operating characteristics are statistical estimates (dependent on the tuning) evaluated through simulation, of the following:

- **Expected number of DLTs** at each dose level
- **Expected number of patients who may be overdosed**
- **Proportion of trials** under which the MTD is correctly identified
- Operating characteristics are **dependent on the tuning parameters**

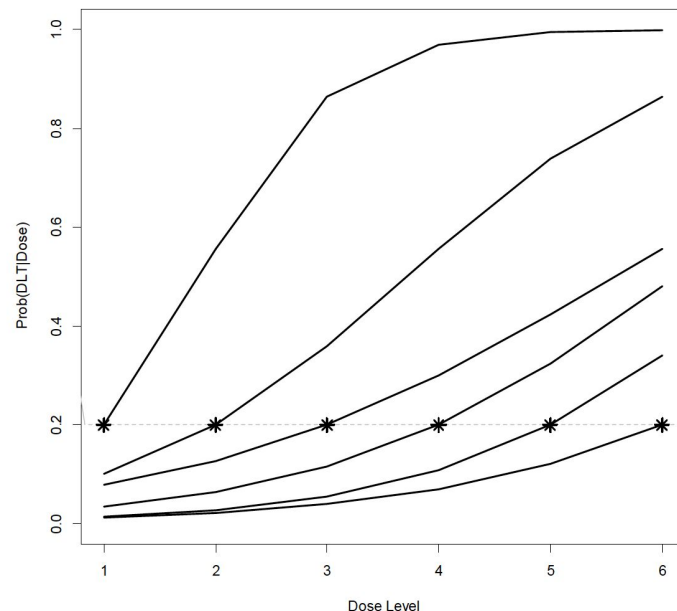


Figure 3 Six true dose-toxicity curves where the true MTD is represented with a star.

Why are simulations important?

- Design performance crucially depends on the TRUE underlying dose-toxicity curve (which we do not know in practice)
- Hence we need to evaluate a design under various assumed true dose-toxicity curves (referred as scenarios), and the performance of the design must be clinically acceptable under all or most scenarios.
- If we observe, under a clinically plausible scenario, that the design is too aggressive, then we need to change the tuning parameters to make the design less aggressive

Phase I Study of Olaparib and Low Dose Thoracic Radiotherapy for Extensive Stage Small Cell Lung Cancer

Clinical parameters

- **Dose limiting toxicities (DLTs)** - any treatment toxicity with a CTCAEv4 Grade ≥ 3 occurring during the study treatment period.
- **MTD** - the dose with toxicity rate that doesn't exceed acceptable threshold of probability of **toxicity of 20%**
- 6 dose levels
- **DLT window** is 3 weeks after completion of radiotherapy
- **Number patients enrolled** - up to 24 patients
- **Cohort size** of 3
- **Recruitment rate** $= 1/30 = 0.03$

Dose level	Olaparib dose
1	50mg BID
2	100mg BID
3	150mg BID
4	200mg BID
5	250mg BID
6	300mg BID

Phase I Study of Olaparib and Low Dose Thoracic Radiotherapy for Extensive Stage Small Cell Lung Cancer

Model Tuning Parameters

- **Prior estimates for the probabilities of DLT** at each dose level:
0.05,0.1,0.2,0.3,0.4,0.5
- Assume **dose-toxicity relationship**:

$$Prob(DLT|dose = x, all\ data) = \left(\frac{\tanh(x) + 1}{2} \right)^a$$

“We note the 3+3 design would not be an appropriate candidate primarily due to the large number of dose levels under investigation (6) and the smaller target toxicity rate (20%) - the 3+3 targets a toxicity rate of 25-33%. Furthermore, there is a rich literature demonstrating the CRM is more accurate in estimating the MTD and safer in reducing the number of DLTs.”

Operating Characteristics (Simulations)

- Prior estimates (from investigator) for the probabilities of a DLT at each dose level are:
 - 0.05, 0.1, 0.2, 0.3, 0.4, 0.5.
- Fill this table using different scenarios on the probability each dose is declared MTD

Operating Characteristics	Dose Level					
	1	2	3	4	5	6
% MTD Selection						
% Subjects Treated						
# Subjects Treated						
Average Toxicity						
True probabilities						

CRM Package: crmsim()

Function Arguments:

- **target** - Target probability of toxicity.
- **prior** - Prior probabilities of toxicity for each dose.
- **true** - True probabilities of toxicity.
- **rate** - Recruitment/accrual rate of subjects in a 30 window
- **cycle** - The length of treatment cycle in days
- **cohort** - Cohort size of subjects entering into the trials.
- **nsubject** - Total number of subjects in one simulation (or trial).
- **model** - Dose-toxicity model (hyperbolic tangent model or logistic model)
- **a0** - Initial value for parameter a. Default is 1.0
- **b** - A fixed parameter for the one-parameter logistic model. Default is 3.0

```
crmsim(target,  
        prior,  
        true,  
        rate,  
        cycle ,  
        cohort = 3,  
        nsubject,  
        model = 1,  
        a0 = 1,  
        seed = 777,  
        nsim = 1000)
```

Modified CRM

In this scenario, the modification is allowing 3 patients in each cohort (non-modified is 1)

Other modifications can include

- Evaluate next before previous patient evaluated
- Skip dose
- Jump a dose level
- And more....

```
crmsim(target,  
        prior,  
        true,  
        rate,  
        cycle ,  
        cohort = 3,  
        nsubject,  
        model = 1,  
        a0 = 1,  
        seed = 777,  
        nsim = 1000)
```


Code Time!



Memorial Sloan Kettering
Cancer Center